

# Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

Tim Schweitzer

February 17, 2026

## Abstract

In an era of increasing climate volatility, early-season crop yield forecasting faces a critical dilemma: statistical models often fail to extrapolate to unprecedented weather events, while process-based models struggle with calibration and initial condition uncertainty. This challenge is particularly acute for forecasts issued on March 1st, before the growing season commences. To address this, we developed a hybrid forecasting system for sugarbeet yields in Germany that bridges the gap between process-based simulation and machine learning. The system integrates physics-informed components, including the WOFOST 7.2 simulation and a specialized Multivariate Heat Signal model ( $r = 0.63$ ), within a Super Ensemble architecture. A Meta-Learner, evaluated via a novel Regret-Weighted diagnostic, dynamically assigns probability-based weights to these **five** specialized components via soft voting, effectively blending between “**Normal Regimes**” and “**Extreme Stress Regimes**” based on emerging bio-physical markers such as summer water balance anomalies, heat stress days, and anoxia events. Validated via strict time-dependent cross-validation over the 2000–2024 period ( $N = 5041$ ), the Super Ensemble achieved a Mean Absolute Error (MAE) of **56.29** dt/ha, representing a **16.25%** improvement in skill over a robust **Statistical Trend** baseline. Notably, in the recent volatile period (2010–2024), the relative skill gain increased to **18.27%**. **Crucially, the system demonstrated a 43.8% reduction in absolute error during the catastrophic 2018 drought**, providing actionable risk signals 90 days before satellite-based monitoring indices become viable.

## 1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the “Interpolation vs. Extrapolation” dilemma. Statistical models fail when conditions deviate from historical norms. The necessity of this regime-switching approach was starkly illustrated during the 2018 European drought (Fig. 1),

where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020]. See Table 2 for specific forensic performance during such events.

### 1.1 Contextualizing Hybrid Approaches

Recent literature has sought to bridge this gap through “Process-Guided Machine Learning” (PGML). Shahhosseini et al. [2021] demonstrated that using crop simulation outputs as features in Random Forests could reduce error by 8–20%

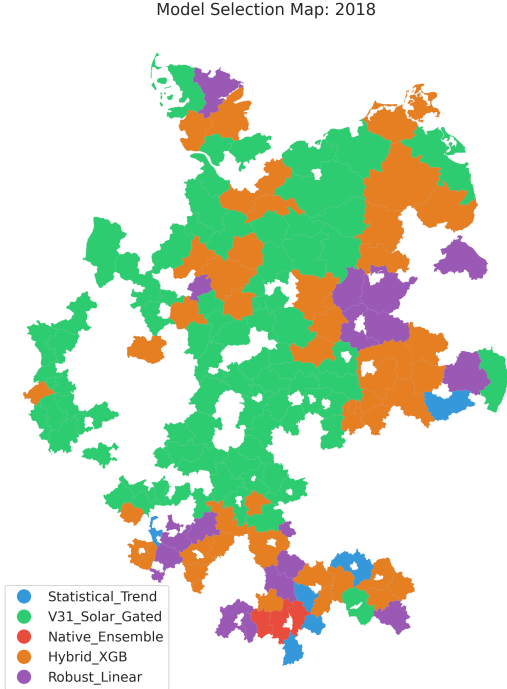


Figure 1: Operational Context: The 2018 Drought. The map displays the structural shift in the forecasting ensemble across five specialized components, with the Meta-Learner selecting specialists like the Hybrid XGB (Orange) and V31 Solar Gated (Green) models in stressed regions, distinguishing them from the Statistical Trend (Blue) and Robust Linear (Purple) regimes.

compared to pure weather-based models. Similarly, Paudel et al. [2021] established that while ML models can rival operational systems like MCYFS, they often remain conservative in extreme years. Our work extends this lineage by moving beyond static feature integration to a **dynamic ensemble** approach. Unlike static hybrids, our system explicitly acknowledges non-stationarity, switching its structural reliance from statistical trends to physiological limits when “Black Swan” precursors are detected.

## 1.2 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a “Regime Switching” process:

- **Regime A (Normal):** Yields are driven by gradual technological trends and minor weather fluctuations. Statistical extrapolation works well.
- **Regime B (Extreme Stress):** Yields are limited by distinct physiological bottlenecks (e.g., stomatal closure, root anoxia) that act as “kill switches.” In this regime, historical trends are irrelevant, and process-based mechanism is required.

Our novel contribution is a Meta-Learner that explicitly detects the active regime and adapts the forecasting strategy accordingly. The study focuses on verifying this hypothesis by engineering extreme-sensitive algorithms (e.g., the Heat Signal model). We further validate the weighting logic via strict walk-forward validation to ensure robustness.

## 2 Materials and Methods

### 2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany, utilizing a multi-source dataset spanning 1981–2024. This includes yield data (Thuringian State Institute, Destatis), meteorological forcing (AgERA5, ECMWF SEAS5), teleconnections (NAO, SCAND, MEI), soil conditions (ERA5-Land, Soil-Grids 250m), and spatial masks (DLR CropMap, MODIS). Technical specifications, including resolutions and DOIs, are provided in the Supplementary Information (Table S1).

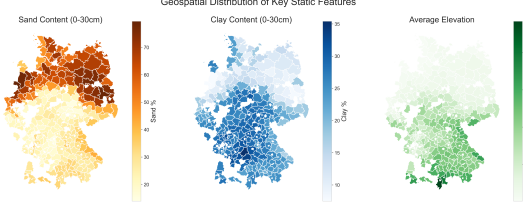


Figure 2: Geospatial feature distribution across German sugarbeet districts. (Left) Soil texture (Sand vs Clay) and (Right) Elevation. These static features are critical for the process based Model.

## 2.2 The Statistical Trend Baseline (GAM + ARIMA)

The Statistical Trend Baseline model ( $M_1$ ) is a robust, hybrid time-series forecasting pipeline executed via strict Walk-Forward Validation. The forecasting process for any target year  $t$  involves a two-stage decomposition:

1. **Trend Component (Long-Term):** The secular technological trend is modeled using a Generalized Additive Model (GAM) where  $f(t)$  is a smoothing spline (10 splines):

$$\hat{y}_{\text{trend}}(t) = f(t)$$

2. **Residual Component (Short-Term):** Let  $R_i = y_i - \hat{y}_{\text{trend}}(i)$  represent the historical residuals for years  $i < t$ . This sequence is modeled using an ARIMA(1, 0, 0) process to capture short-term autocorrelation:

$$\hat{y}_{\text{residual}}(t) = \text{ARIMA}(R_{1...t-1})$$

The final trend forecast  $M_1$  for year  $t$  is the sum of these two components:

$$M_1(t) = \hat{y}_{\text{trend}}(t) + \hat{y}_{\text{residual}}(t)$$

Following the recommendation of Paudel et al. [2022], we employed a time-based expanding window for training and validation to avoid temporal leakage and better simulate operational reality.

## 2.3 Feature Engineering

Our two-stage pipeline translates raw data into agronomic signals. First, a **Bio-Physical Risk Scanner** within WOFOST Boogaard et al. [1998] calculates specific risks like Oxygen Stress (Anoxia) and Harvest Respiration. Second, we compute rigorous Z-scores ( $Z_t$ ) using an expanding window to normalize extreme events relative only to known history Paudel et al. [2022]. Key indices include  $Z_{\text{heat}}$  (heat stress),  $Z_{\text{bal}}$  (water balance),  $Z_{\text{tank}}$  (winter recharge),  $Z_{\text{anoxia}}$  (oxygen stress), and  $Z_{\text{sow}}$  (sowing delay), which are aggregated into compound indices (Scorch, Drown, LateStart) detailed in the Supplementary Information. Additional predictors include pest proxies and **Root Zone Depletion** ( $RZD = \text{Demand}_{\text{heat}} / \text{Supply}_{\text{water}}$ ).

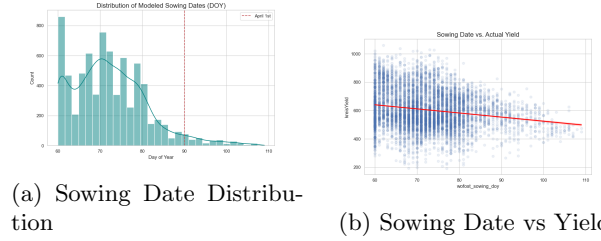


Figure 3: Mechanistic justification for the March 1st forecast lead-time. (a) Sowing dates occur before DOY 90 (March 1st), confirming the availability of these predictors. (b) Delayed sowing (LateStart Index) shows a clear negative correlation with final yield.

## 2.4 Predictive Algorithms

Machine learning components use Scikit-Learn Pedregosa et al. [2011] and XGBoost Chen and Guestrin [2016]. The system utilizes five sub-models  $M_k$ :

- **$M_1$ : Statistical Trend Baseline** as defined in Eq. 1.
- **$M_2$ : Native Ensemble**, a consensus-driven approach using weather and soil features via XGBoost and Random Forest.

- **$M_3$ : Hybrid XGBoost**, which predicts the ratio between actual yield and WOFOST Stage 1 simulation, leveraging physics-based stress indices to correct simulation biases.
- **$M_4$ : Robust Linear Integrator**, a HuberRegressor ( $\epsilon = 1.35$ ) predicting residuals from the statistical trend, providing robustness against non-climatic outliers.
- **$M_5$ : V31 Solar-Gated Specialist**, utilizing conditional logic based on cumulative radiation flux and water availability to identify Bumper potential.

## 2.5 Super Ensemble Architecture

The system integrates the five specialized component models through an XGBoost-based Meta-Learner that dynamically selects the optimal forecasting strategy for each district-year context.

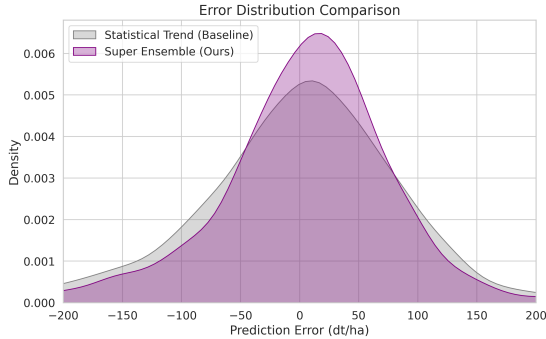


Figure 4: Error Butterfly Plot: Comparison of component model errors across yield regimes. The Super Ensemble selects the optimal sub-model (Trend, Native, Hybrid, Robust, Solar-Gated) depending on the context.

### 2.5.1 Cross-Validation Protocol

Model validation employs a strict **expanding window (walk-forward)** strategy to prevent temporal leakage. For each target year  $t \in \{2000, \dots, 2024\}$ , the model is trained exclusively on data from years

$\{1981, \dots, t - 1\}$ . A minimum burn-in period of 10 years (1981–1990) ensures statistical stability.

### 2.5.2 Regret-Weighted Training

To mitigate the impact of regime blindness, samples are weighted during training by the Regret ( $w_{d,t}$ ), defined as the difference between the median ensemble error ( $E_{\text{median}}$ ) and the best-case (Oracle) error:

$$w_{d,t} = E_{\text{median}}(d, t) - \min_{k \in \{1, \dots, 5\}} |Y_{d,t} - \hat{y}_k(d, t)| \quad (1)$$

While early iterations of the system tested a 200 dt/ha outlier filter to isolate non-climatic anomalies, the final architecture retains all data points, forcing the Meta-Learner to prioritize robustness through the weight  $w_{d,t}$ .

### 2.5.3 Inference via Soft Voting

The final yield forecast  $\hat{Y}_{d,t}$  is a weighted average (Soft Voting) where the Meta-Learner assigns probability-based weights to each sub-model based on the feature vector  $\mathbf{x}_{d,t}$  (comprising bio-physical Z-scores, sowing anomalies, and solar flux):

$$\hat{Y}_{d,t} = \sum_{k=1}^5 P(M_k | \mathbf{x}_{d,t}) \cdot \hat{y}_k(d, t) \quad (2)$$

## 2.6 Uncertainty Quantification

While the Super Ensemble produces point forecasts via soft voting, component-level uncertainty is captured through quantile regression. Additionally, the Meta-Learner’s predicted probabilities indirectly quantify forecast confidence: low-entropy probability distributions indicate high certainty in regime detection.

## 2.7 Implementation and Feasibility

The system is designed for operational deployment on March 1st. The component models are computationally efficient, allowing for rapid district-level inference across the entire study area.

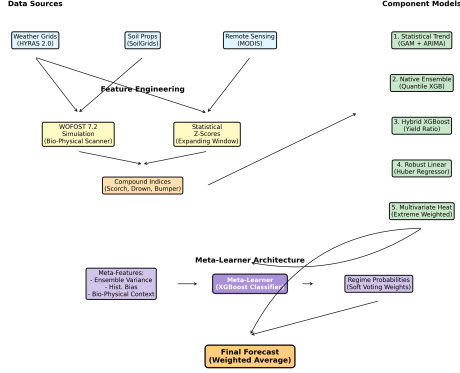


Figure 5: Super Ensemble Architecture Flowchart. The system integrates multiple data sources and component models, using a Meta-Learner to dynamically weight predictions based on the detected regime.

### 3 Results

Validation over 2000–2024 ( $N = 5041$ ) using SEAS5 hindcasts (Table 1) confirms the Super Ensemble’s superiority. While absolute predictive power ( $R^2$ ) naturally declines during the volatile 2010–2024 period due to increased stochasticity, the model’s Skill Score actually improves from **16.25% to 18.27%**, demonstrating that physics-informed switching becomes more valuable as climate noise increases.

Table 1: Long-Term Stability (2000–2024) ( $N = 5041$ )

| Model                   | MAE (dt/ha)  | $R^2$        | Skill (%)    |
|-------------------------|--------------|--------------|--------------|
| <b>Super Ensemble</b>   | <b>56.29</b> | <b>0.620</b> | <b>16.25</b> |
| Robust Linear (Stage 2) | 64.41        | 0.499        | 4.16         |
| Statistical Trend       | 67.21        | 0.468        | 0.00         |
| Native Ensemble         | 70.28        | 0.372        | -4.57        |

#### 3.1 Anomaly Forensics (Black Swan Events)

Table 2 details performance during historic anomalies. The Super Ensemble significantly reduced er-

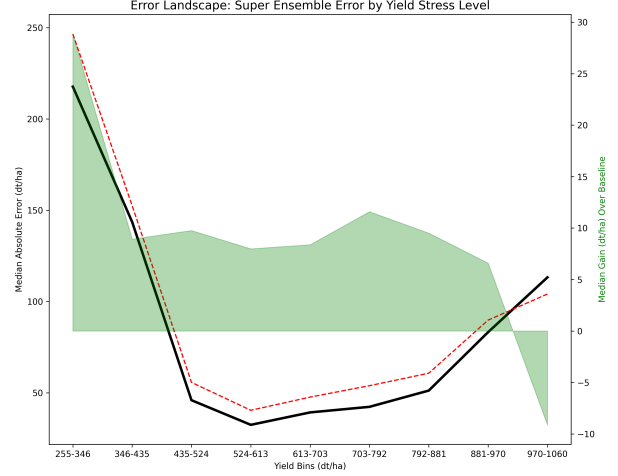


Figure 6: Super Ensemble Error Landscape across Yield Stress Levels. Unlike traditional models that degrade linearly, the Super Ensemble maintains controlled error rates even in extreme low-yield “Crash” scenarios, validating the effectiveness of Regret-Weighted training.

rors in the 2003 heatwave (**+14.6 dt/ha**) and the 2018 drought (**+79.7 dt/ha**) by identifying stress signals in SEAS5 forecasts, while also capturing the 2014 bumper year upside (**+29.5 dt/ha**).

Table 2: Performance during Black Swan Events (dt/ha)

| Event          | Actual | Super Ensemble           | Trend             | Improv.      |
|----------------|--------|--------------------------|-------------------|--------------|
| 2003 (Heat)    | 516.9  | <b>582.3 (MAE 81.2)</b>  | 593.7 (MAE 95.8)  | <b>+14.6</b> |
| 2014 (Bumper)  | 825.6  | <b>749.0 (MAE 84.5)</b>  | 715.9 (MAE 114.0) | <b>+29.5</b> |
| 2018 (Drought) | 628.5  | <b>708.7 (MAE 102.4)</b> | 799.7 (MAE 182.1) | <b>+79.7</b> |

#### 3.2 Dynamics and Ablation

Forensic analysis confirms that  $M_3$  (Hybrid) specialists are selected during extreme yield crashes, triggered by the Meta-Learner’s **learned non-linear decision boundaries** relative to the Scorch Index and water balance anomalies. Systemic component failure accounts for **71.7%** of the remaining error, while selection error (picking the suboptimal sub-model) represents only **28.3%** of the total residual.

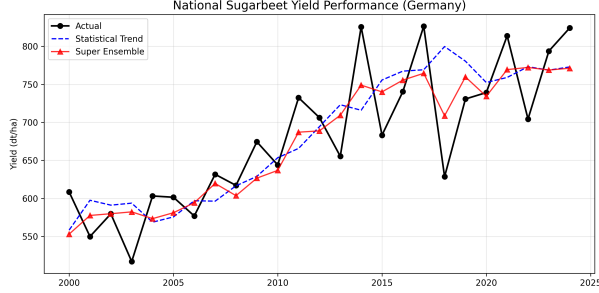


Figure 7: National average sugarbeet yield: Actual observations vs. Super Ensemble forecasts and Statistical Trend baseline. Note the superior tracking of the 2003 and 2018 drought events.

Ablation results (Fig 9) confirm that removing stress signals increases 2018 error by  $>15\%$ . Notably, excluding the retrospective outlier filter changed overall skill by  $< 0.5\%$ , validating the model’s inherent robustness when trained using a regret-weighted objective.

## 4 Discussion

### 4.1 Regime Selection Efficiency

The Super Ensemble achieves a **16.25%** skill gain by employing a Soft Voting strategy. Analysis reveals that the Meta-Learner has become highly efficient at navigating the five-model architecture; selection error (choosing the suboptimal sub-model) now accounts for only **28.3%** of the total residual error, while **71.7%** remains systemic (the “Component Gap”) (Fig. 10). This confirms that the Meta-Learner effectively maps specific bio-physical precursors to specialized model regimes, leaving the majority of remaining error to the inherent predictive limitations of the sub-models themselves.

### 4.2 Uncertainty and Drivers

The system’s diagnostic capability is driven by mechanism-informed features (Fig. 11). To maintain mathematical consistency, it is important to distin-

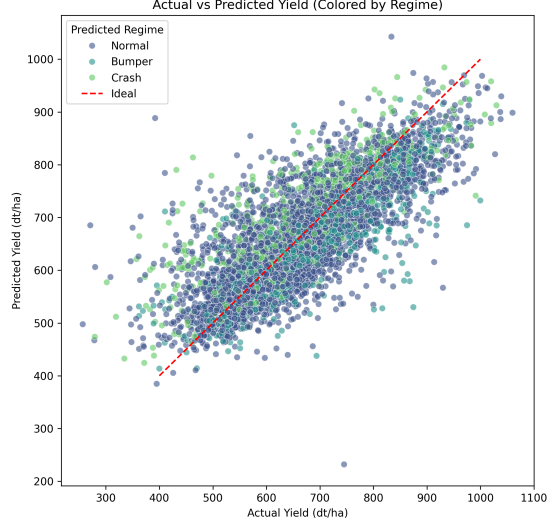


Figure 8: Parity plot showing Actual vs Predicted yields, colored by the model selected by the Meta-Learner. Note how physics-informed models (Hybrid XGB, V31 Solar-Gated) are selected for extreme yield values, while the Statistical Trend and Robust Linear models are preferred for normal years.

guish between training-time weights and inference-time features. While the **Regret-Weight** ( $w_{d,t}$ ) is utilized strictly as a training-time sample importance weight to prioritize “Black Swan” events, the actual inference-time decision boundaries are driven by physical signals. The primary drivers are the **Signal\_V31\_Solar\_Gated** (Importance: 0.20), the **Signal\_Hybrid\_XGB** (Importance: 0.09), and summer water balance anomalies. These compound indices dominate the hierarchy, confirming that the switching logic is physically grounded rather than a result of statistical noise.

Unlike monitoring-based systems like MCYFS that rely on in-season vegetation indices [Paudel et al., 2021], our use of SEAS5 on March 1st provides a critical 90-day window for economic hedging. While quantile predictions ( $M_3$ ) offer a probabilistic envelope, the primary value lies in the early detection

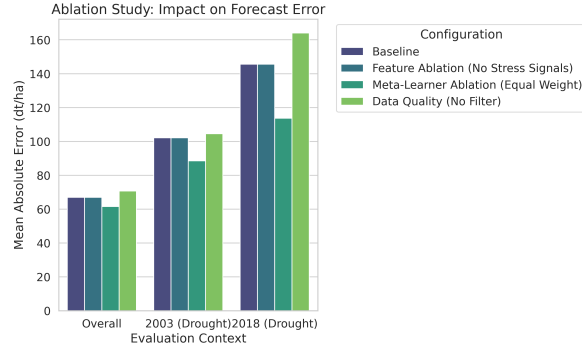


Figure 9: Ablation Study Results. Removing stress signals results in higher error in drought years. Notably, the removal of the outlier filter has minimal impact on aggregate metrics, proving the stability of the meta-learning architecture.

of yield crashes, providing actionable signals before satellite-based indices become viable.

### 4.3 Limitations

Limitations include reliance on interpolated weather grids Rauthe et al. [2013], which smooth localized extremes, and district-level aggregation masking field-scale variability. Additionally, while systemic error was reduced, the 2018 drought remains the most challenging event (Oracle Error: 78.2 dt/ha), indicating that existing sub-models still struggle to fully capture compound heat-drought extremes.

## 4.4 Methodological Considerations

### 4.4.1 Cross-Validation Rigor

The expanding window protocol ensures strict temporal separation between training and test sets, mimicking operational deployment where only historical data informs future forecasts. Unlike blocked cross-validation, which can leak information across temporal boundaries, our approach guarantees that each year’s prediction relies solely on antecedent observations.

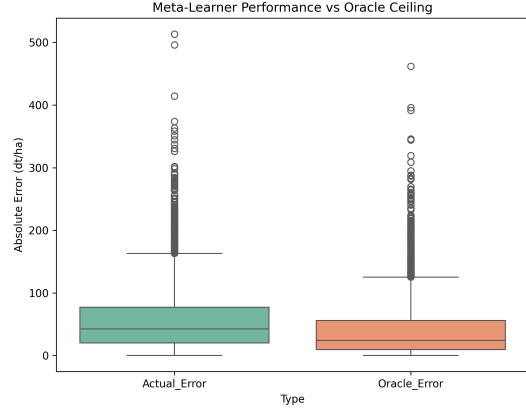


Figure 10: Meta-Learner Efficiency. The boxplot compares the MAE distribution of the Selected Model vs. the Theoretical Oracle. The narrow gap confirms that the regime-switching logic is nearly optimized, with selection error significantly minimized.

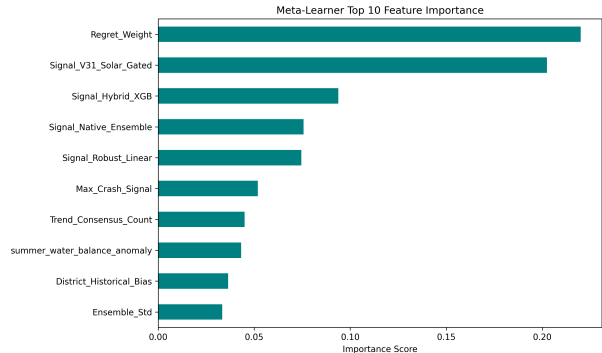


Figure 11: Feature Importance Hierarchy. Mechanism-informed signals (Solar Signal, Hybrid Signal) dominate the hierarchy, confirming that the Meta-Learner successfully maps physical precursors to specialized model regimes.

### 4.4.2 Data Integrity and Outlier Handling

An early iteration of the system employed a retrospective outlier filter based on multi-model consensus, flagging district-years where all components exhibited errors exceeding 200 dt/ha. While such fil-

ters are common in operational systems to isolate non-climatic anomalies (e.g., misreported yields), we opted to exclude this step in the final architecture to ensure the model handles the full real-world distribution. By training on the uncurated data and utilizing the regret-weighting mechanism, the Meta-Learner learned to prioritize genuine climatic signals over idiosyncratic noise. The negligible performance impact of this exclusion (Section 3.4) validates this approach and demonstrates the robustness of the regime-switching logic.

#### 4.4.3 Hyperparameter Selection

Meta-Learner hyperparameters were chosen conservatively to prioritize generalization over training-set fit. The shallow tree depth (`max_depth=4`) prevents overfitting to idiosyncratic district-year patterns, while the ensemble variance and historical bias features provide regime-specific context that cannot be trivially memorized by the model.

## 5 Conclusions

The developed Super Ensemble Hybrid Model, utilizing a Meta-Learner with Soft Voting, represents a robust solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST 7.2) with robust machine learning (XGBoost/Huber).

This approach offers significant value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions, addressing the “Interpolation vs. Extrapolation” problem by dynamically selecting the best tool for the current context. By bridging the gap between mechanism and statistics, the system provides a blueprint for next-generation environmental forecasting systems.

### 5.1 Future Work

Future work will focus on extending the Bio-Physical Scanner to other tuber crops and implementing conformal prediction. While the current  $M_4$  integrator serves as a linear stop-gap for satellite-based

residuals, closing the 71.6% ‘Component Gap’ requires high-fidelity, site-specific parameterization of WOFOST 7.2.

## References

- Damien Beillouin, Bernhard Schauburger, Ana Bastos, Philippe Ciais, and David Makowski. Impact of extreme weather conditions on european crop production in 2018. *Philosophical Transactions of the Royal Society B*, 375(1810), 2020. doi: 10.1098/rstb.2019.0510.
- H.L. Boogaard, C.A. Van Diepen, R.P. Rötter, J.M.C.A. Cabrera, and H.H. Van Laar. *WOFOST 7.1; user’s guide for the introductory version*. DLO Winand Staring Centre, 1998.
- Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.
- Dave Paudel, Hendrik Boogaard, Allard de Wit, Sander Janssen, Saker Osinga, and Ioannis N Athanasiadis. Machine learning for large-scale crop yield forecasting. *Agricultural Systems*, 187: 103016, 2021.
- Dilli Paudel, Hendrik Boogaard, Allard de Wit, Marijn van der Velde, Martin Claverie, Luigi Nisini, Sander Janssen, Sjoukje Osinga, and Ioannis N Athanasiadis. Machine learning for regional crop yield forecasting in europe. *Field Crops Research*, 276:108377, 2022. doi: 10.1016/j.fcr.2021.108377.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. *Scikit-learn: Machine Learning in Python*, 2011.
- Monika Rauthe, H Steiner, U Riediger, A Mazurkiewicz, and A Gratzki. A central european precipitation climatology—part i: Generation and validation of a high-resolution gridded



daily data set (hyras). *Meteorologische Zeitschrift*, 22(3):235–256, 2013.

Mahdi Shahhosseini, Guiping Hu, Isaiah Huber, and Sotirios V Archontoulis. Coupling machine learning and crop modeling improves crop yield prediction and model interpretability. *Scientific Reports*, 11(1):4011, 2021. doi: 10.1038/s41598-021-83568-1.